Environment benchmark: masked correctness and factual consistency

Percent of all questions in each set (set A: 141, set B: 272, set C: 272) by the judge panel's voted label. Factual consistency: the labels sum to 100%. Masked correctness: same category (incl. exact), different category, not comparable, and no label sum to 100%; exact is a part of same category.† "–": the configuration did not answer that set.

Environment configuration	Masked correctness (set A)					Factual consistency (set B)			Masked correctness (set C)					Factual consistency (set C)			Environment cost and time	
	exact	same category (incl. exact)†	different category	not comparable	no label*	consistent	inconsistent	no label*	exact	same category (incl. exact)†	different category	not comparable	no label*	consistent	inconsistent	no label*	Total cost (USD)	Average answer time (s)
1. Current (Europe PMC + LitSense, without changes 1–4)	42.6%	87.2%	12.8%	0.0%	0.0%	95.2%	0.0%	4.8%	–	–	–	–	–	–	–	–	$0.443	86.1
2. Current with changes 1–4	36.9%	82.3%	17.0%	0.0%	0.7%	97.1%	0.4%	2.6%	–	–	–	–	–	–	–	–	$0.416	94.3
3. OpenRouter search: Exa instant, DeepSeek v4 Flash	42.6%	85.8%	13.5%	0.7%	0.0%	96.0%	0.0%	4.0%	–	–	–	–	–	–	–	–	$3.258	74.2
4. OpenRouter search: Parallel basic, DeepSeek v4 Flash	33.3%	80.9%	19.1%	0.0%	0.0%	96.3%	0.4%	3.3%	–	–	–	–	–	–	–	–	$2.516	84.2
5. OpenRouter search: Perplexity, DeepSeek v4 Flash	39.0%	84.4%	14.9%	0.7%	0.0%	**97.4%**	0.0%	2.6%	–	–	–	–	–	–	–	–	$2.470	78.8
6. OpenRouter search: Google (native), Gemini 3.1 Flash Lite	41.1%	83.0%	16.3%	0.7%	0.0%	96.7%	0.0%	3.3%	–	–	–	–	–	–	–	–	$21.086	82.4
7. OpenRouter search: OpenAI (native), GPT-6-luna	41.8%	85.1%	14.9%	0.0%	0.0%	95.6%	0.0%	4.4%	–	–	–	–	–	–	–	–	$10.800	79.9
8. Case information only, no retrieval (DeepSeek v4 Flash 0731)	–	–	–	–	–	–	–	–	98.9%	99.6%	0.4%	0.0%	0.0%	100.0%	0.0%	0.0%	$0.026	7.7
9. Gemini 3 Flash knowledge, no search (gemini-3-flash-preview)	**43.3%**	86.5%	13.5%	0.0%	0.0%	96.7%	0.0%	3.3%	–	–	–	–	–	–	–	–	$0.660	72.8

- Questions from the 272 cases in cases/combined_272_whole_chunking.json: set A (141), a value hidden from the case; set B (272), a value the case never states; set C (272), information the case states, written by an LLM (google/gemini-3.5-flash-lite). Configurations 1-7 and 9 answered the same sets A and B; configuration 8 answered set C.
- Judge panel: google/gemini-3.5-flash-lite, qwen/qwen3.8-flash, openai/gpt-6-luna. Each answer's label is the majority vote; with no majority, the first (main) judge decides. Set C answers are graded by both metrics. Judging cost $3.30 in total, not included in the costs above.
- * No label: medsim gave no answer (it found no usable literature, or for set C the case did not answer the question), the run failed, or there was no judge verdict; it is not a judge label.
- Total cost: every environment component for all questions of the configuration (medsim Stages A–C, search fees, and the search model's tokens), from OpenRouter's usage.cost. Average answer time: mean wall time per question, all components included.
- Configurations 6 and 7 use the model provider's own (native) search. Their documents are the search model's quotations or summaries of each cited source, not page excerpts as returned by Exa, Parallel, and Perplexity.
- Configuration 8 has retrieval switched off: medsim answers from the case information, or not at all, with deepseek/deepseek-v4-flash-0731 for every stage.
- Configuration 9 replaces the search with google/gemini-3-flash-preview, given the question, the case information as medsim sees it, and the diagnosis, and asked for the value that typically fits; its answer is the one document medsim's stages use.
- Since 2026-09-29, when the documents give no usable basis, medsim's synthesizer makes a best guess from the case and diagnosis instead of declining. The answers configurations 1-7 had declined were replayed with it on the same documents (no new search).
- Masked correctness labels: exact (the answer matches the true value), same category (same low/normal/high category, or the same finding with a minor detail wrong), different category, not comparable.
- † Same category (incl. exact) = exact + same category: the share of answers at least in the true value's category. Exact is a part of it: same category, different category, not comparable, and no label sum to 100%; same category alone is this column minus exact.